

ROS2 ATOSFleetManagement Cheatsheet

Supported native installations

Ubuntu 20.04 -> ROS 2 Foxy
Ubuntu 22.04 -> ROS 2 Humble
Ubuntu 24.04 -> ROS 2 Jazzy

Always source ROS before building or launching

Ubuntu 24.04: source /opt/ros/jazzy/setup.bash
Ubuntu 22.04: source /opt/ros/humble/setup.bash
Ubuntu 20.04: source /opt/ros/foxy/setup.bash
Check: echo \$ROS_DISTRO

Local workspace build (manual)

```
cd <ATOS_WORKSPACE>
source /opt/ros/<jazzy|humble|foxy>/setup.bash
colcon build --symlink-install --cmake-args -DWITH_TRUCK_OBJECT_CONTROL=ON
source install/setup.bash
```

FleetManagement helper build

```
cd <ATOS_REPO_ROOT>
source /opt/ros/<jazzy|humble|foxy>/setup.bash
./scripts/build_atosfleetmanagement.sh
```

Local launch

```
source /opt/ros/<jazzy|humble|foxy>/setup.bash
source <ATOS_WORKSPACE>/install/setup.bash
ros2 launch atos launch_atosfleetmanagement.py foxbridge:=False insecure:=True
TruckObjectGUI: http://localhost:8420
```

Docker launch

```
docker compose -f docker-compose-fleetmanagement.yml up --build
WITH_TRUCK_SIMULATOR=True docker compose -f docker-compose-fleetmanagement.yml
  up --build
WITH_TRUCK_SIMULATOR=False docker compose -f
  docker-compose-fleetmanagement.yml up --build
```

Service / deployment

```
sudo cp scripts/atosfleetmanagement.env.example
  /etc/default/atosfleetmanagement
sudo cp scripts/atosfleetmanagement.service
  /etc/systemd/system/atosfleetmanagement.service
sudo systemctl daemon-reload
sudo systemctl enable --now atosfleetmanagement
```

Main runtime pieces

```
launch file:      ros2 launch atos launch_atosfleetmanagement.py
build script:     ./scripts/build_atosfleetmanagement.sh
run script:       ./scripts/run_atosfleetmanagement.sh
compose file:     docker-compose-fleetmanagement.yml
route asset:      conf/conf/RuralRoad_center_of_driving_lane_ccw.geojson
```

ROS interfaces

Send truck input to TruckObjectControl:
/atos/truck_objects/cot
Current temporary payload format:

id=<truck_id>;distance_m=<value>;tcp_connected=<0|1>

Read commanded speed from:

/atos/truck_objects/speed_command

Read live truck state for GUI/map from:

/atos/truck_objects/state

Useful inspection commands:

ros2 topic list

ros2 topic echo /atos/truck_objects/state

ros2 topic echo /atos/truck_objects/speed_command